



frozen (28.7M)



trainable (10.98M)



last 4 frames  
640×400 @ 100 ms  
(300 ms span)



### DINOv3 ViT-S+/16

frozen \*  
28.7M params, bf16



### 41 tokens × 384

per frame:  
CLS + 5×8 patch sample

*+ learned spatial &  
temporal embeddings*



### Transformer trunk

6 layers, d=384, 6 heads  
164 tokens, bidirectional

last-frame  
CLS readout

### held-state head

36 Bernoulli logits  
sample @ T=1 → diff  
→ key events

### mouse head

tanh dx, dy  
±500 / ±250 counts